

CS & IT ENGINEERING

DISCRETE MATHS
SET THEORY



Lecture No. 5



By- SATISH YADAV SIR

TOPICS TO BE COVERED

01 Composition of Function

02 Theorems in Composition
of function

03 Examples in Composition
of function

Functions

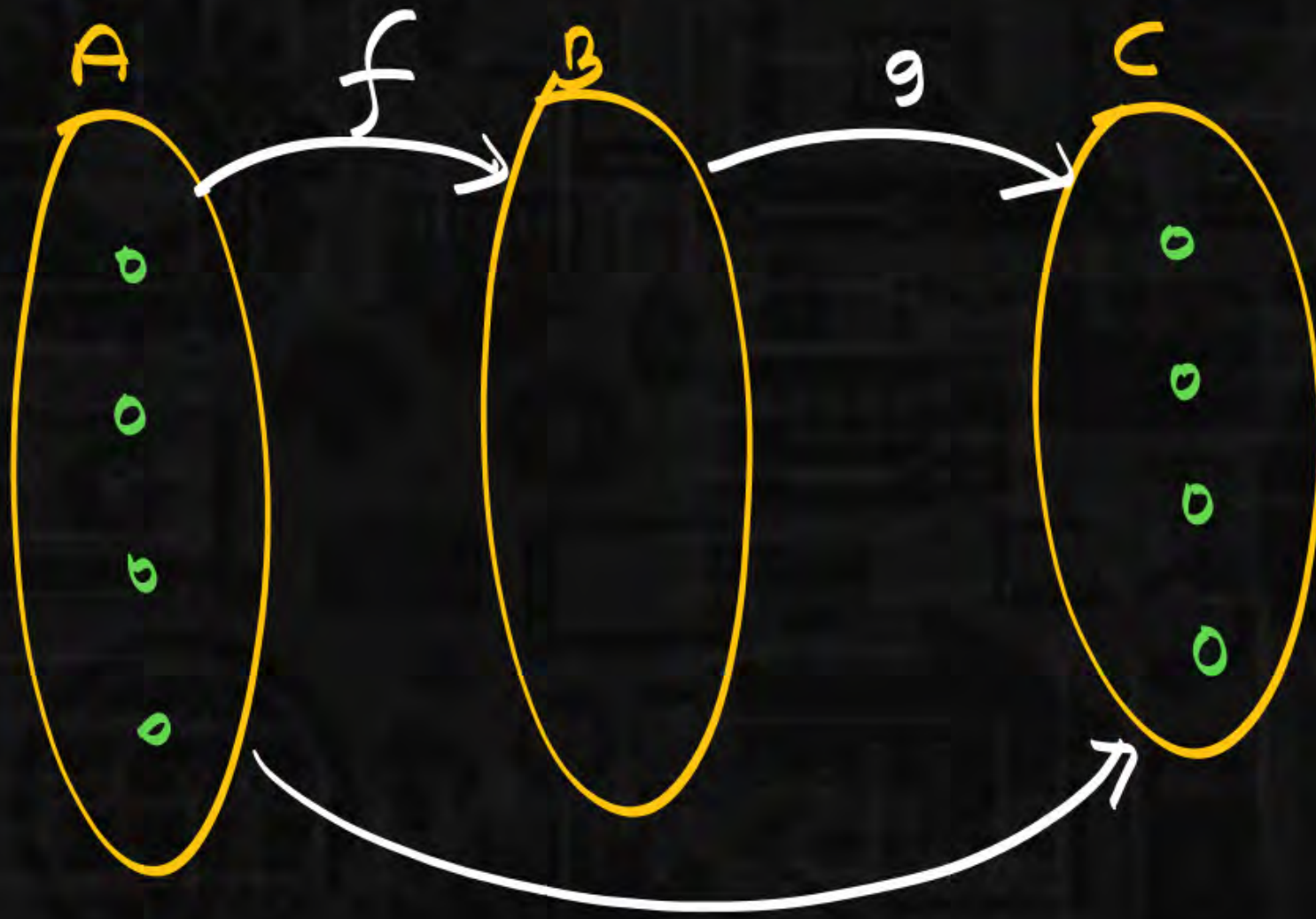


$$f: A \rightarrow B$$

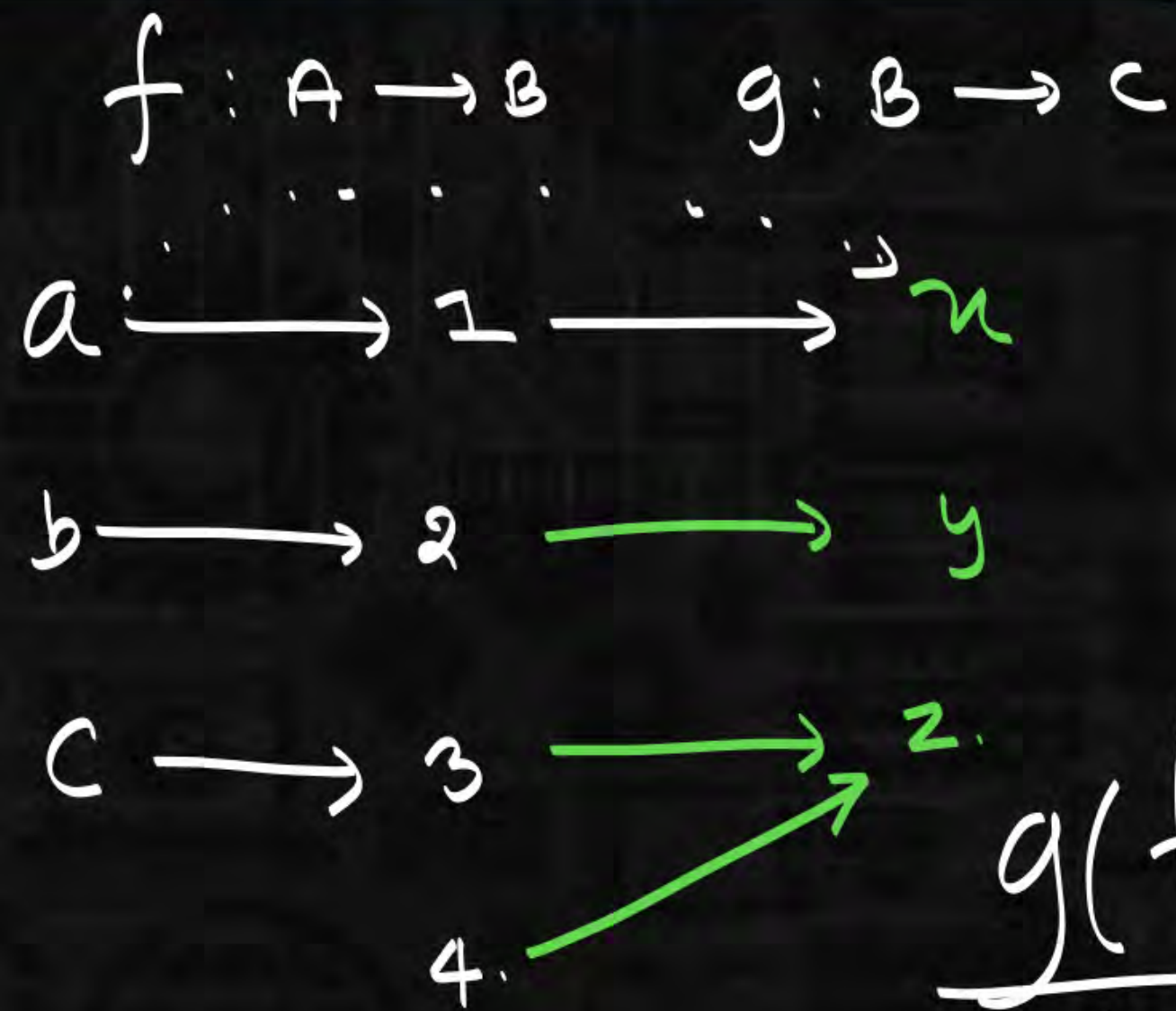
$$g: B \rightarrow C$$

composition of function: $A \rightarrow C$.

$$g \circ f: A \rightarrow C$$



Functions



$$f: A \rightarrow B$$

$$f(a) = 1$$

$$f(b) = 2$$

$$f(c) = 3$$

$$g: B \rightarrow C$$

$$g(1) = x, \quad g(2) = y$$

$$g(f(a)) = x$$

$$g(2) = y$$

$$g(f(b)) = y$$

$$\frac{g(f(\underline{c}))}{\underline{g \circ f}}$$

Functions

$$\begin{array}{c}
 g(f(x)) \\
 \downarrow \\
 \underline{g \circ f}
 \end{array}$$

Functions



$$f\left(\frac{n}{2}\right) = \frac{n+1}{2}$$

$$g \circ f = g(\underline{f(n)})$$

$$= g(\underline{n+1})$$

$$= 2(n+1)+3$$

$$2n+2+3 = 2n+5$$

$$g(\underline{n}) = 2\underline{n}+3$$

$$\boxed{g \circ f \neq f \circ g}$$

$$f \circ g = f(\underline{g(n)})$$

$$= f(2n+3)$$

$$= 2n+3+1$$

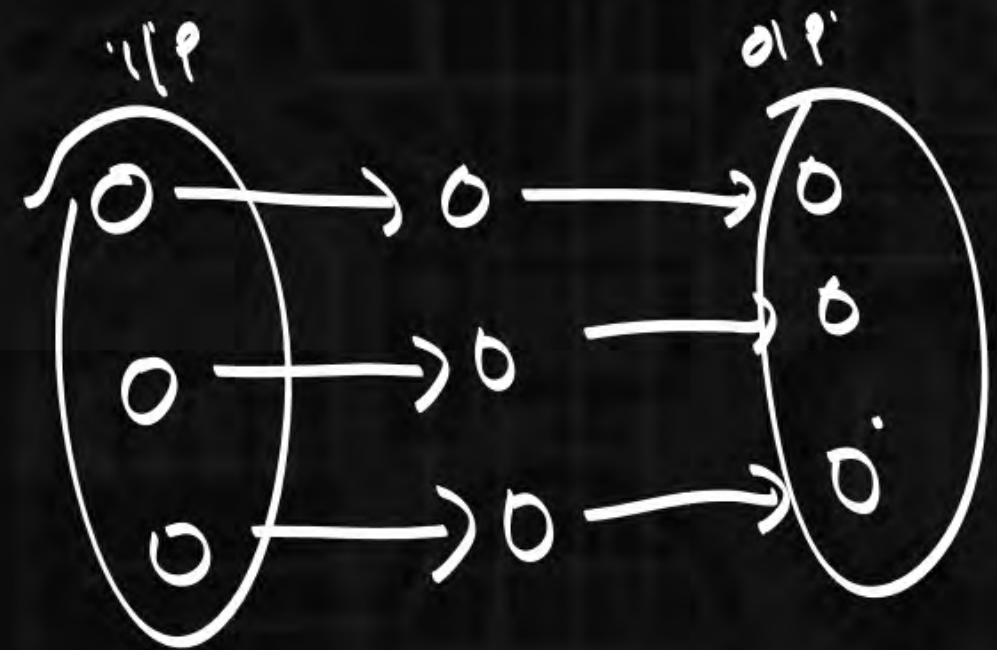
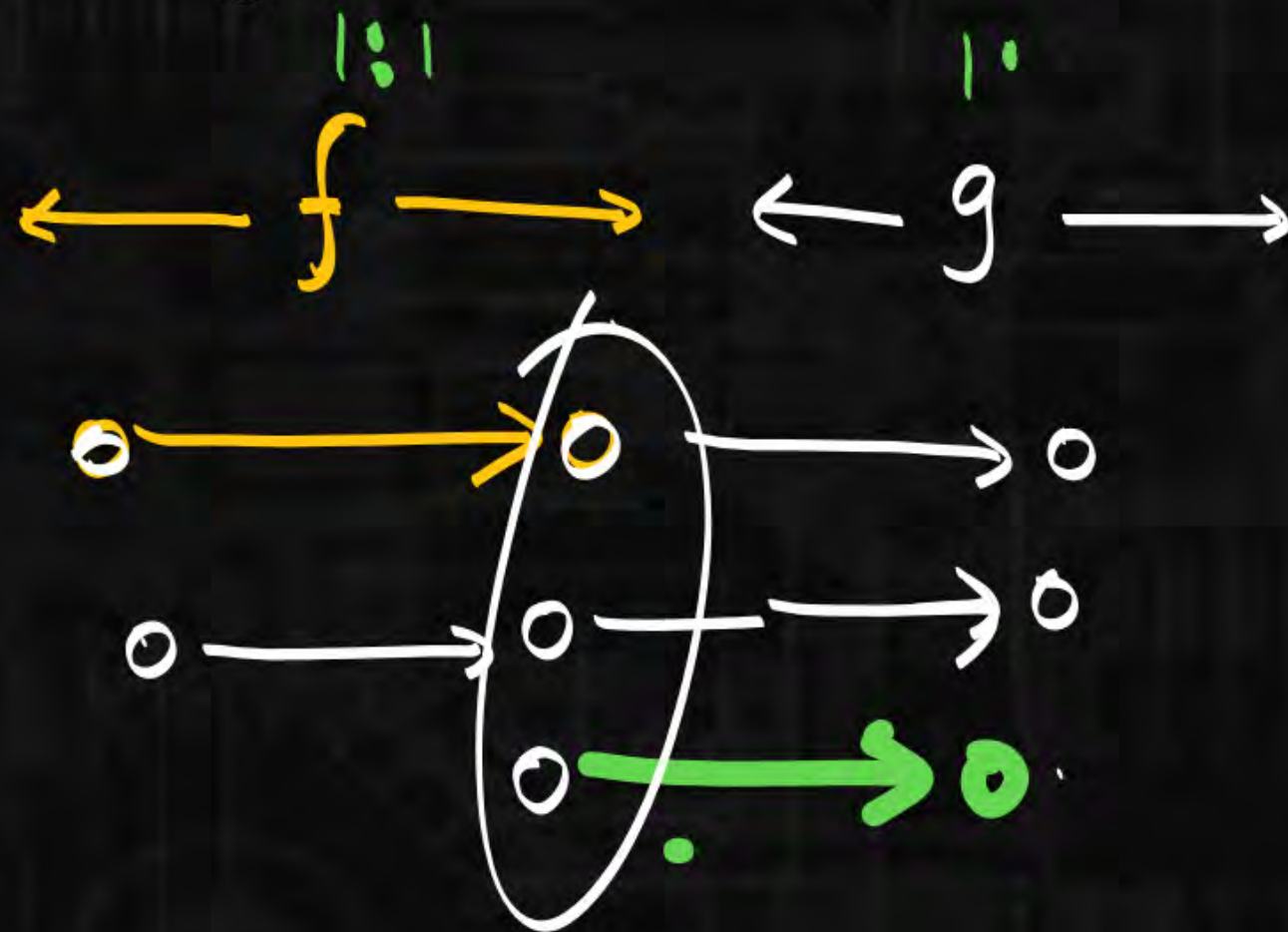
$$= 2n+4$$

Functions



$$f: A \rightarrow B \quad g: B \rightarrow C$$

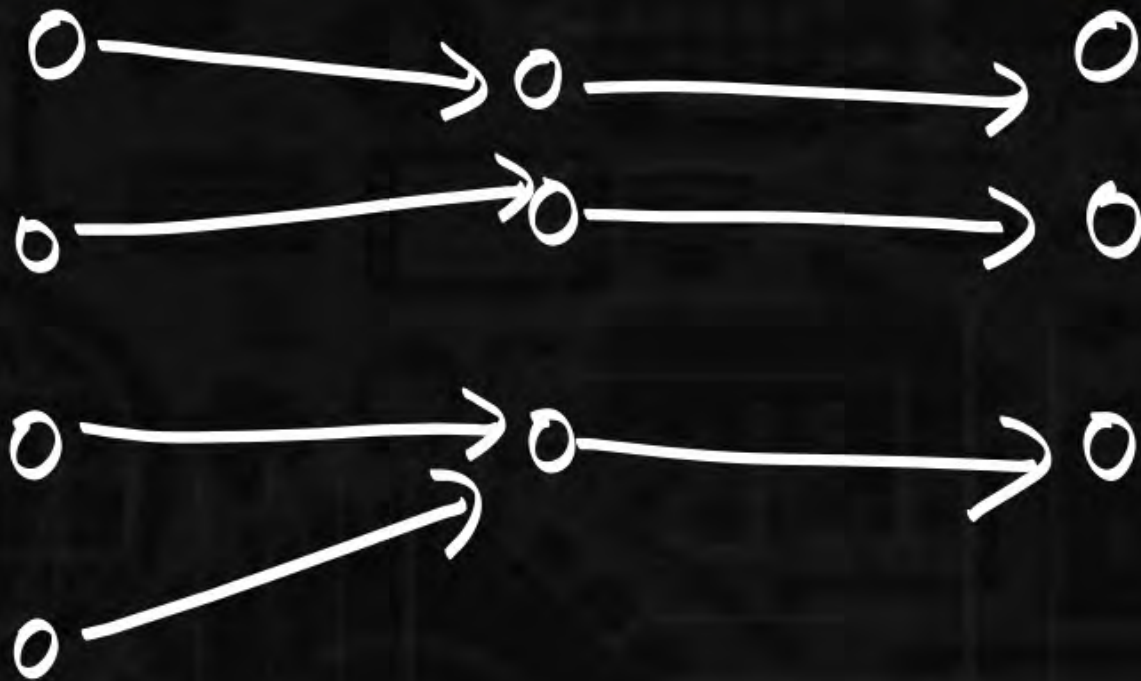
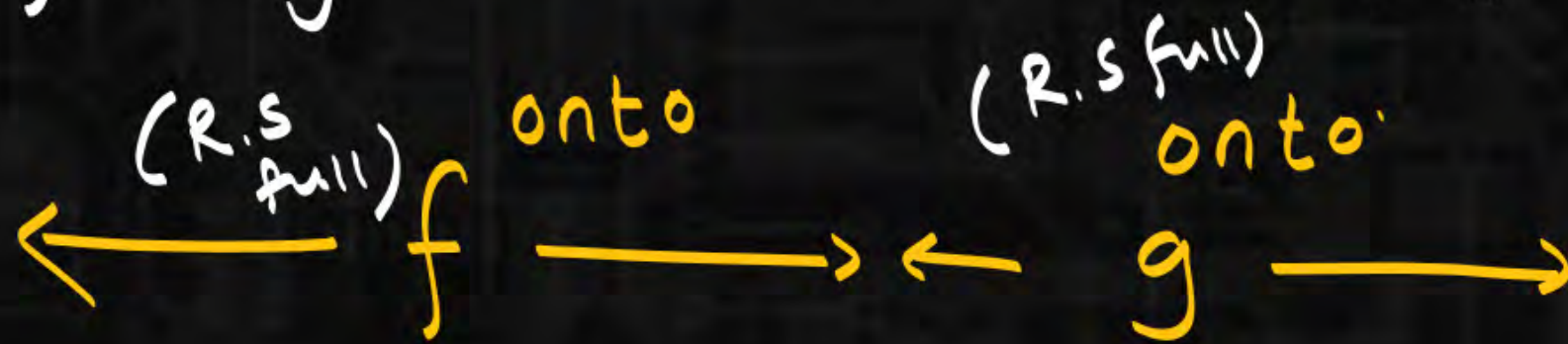
1) if f & g are 1:1 function then $g \circ f$ is also 1:1.



Functions



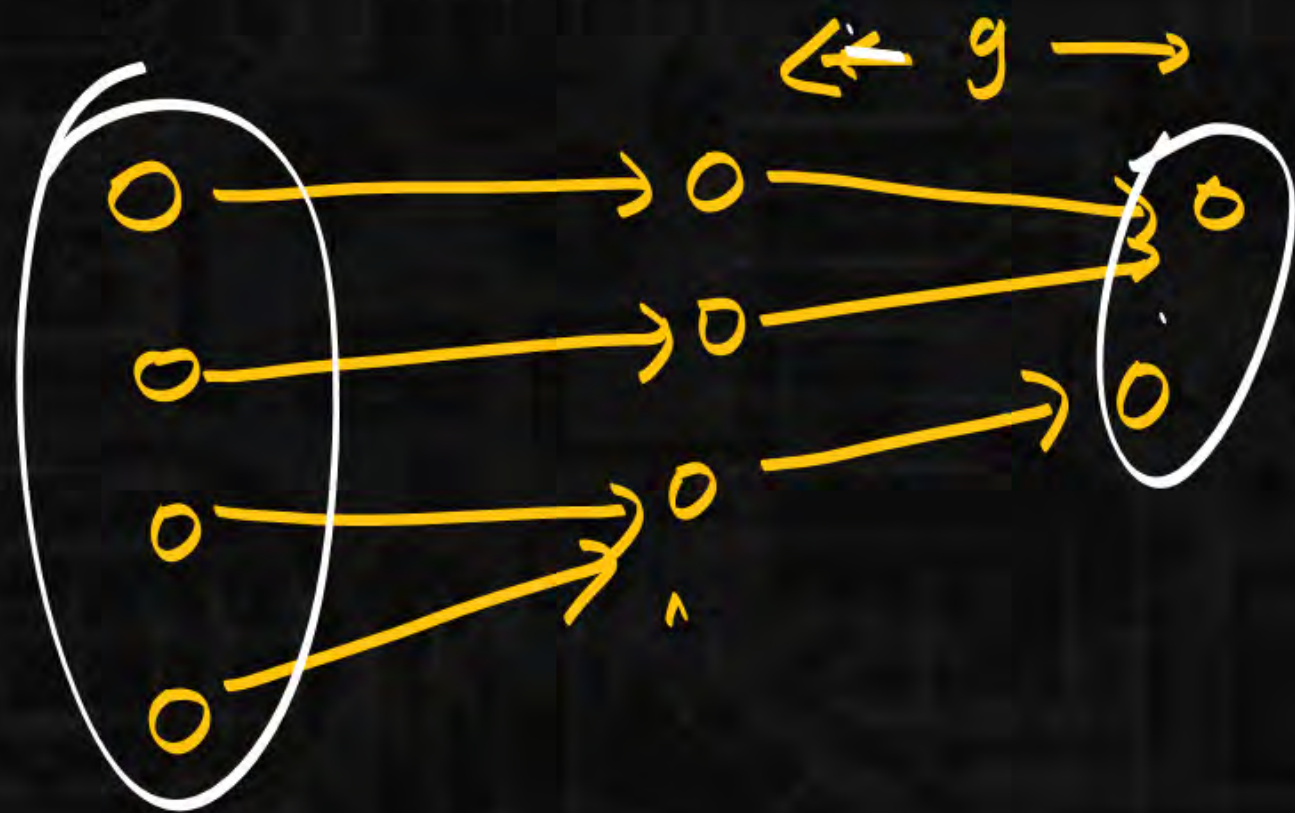
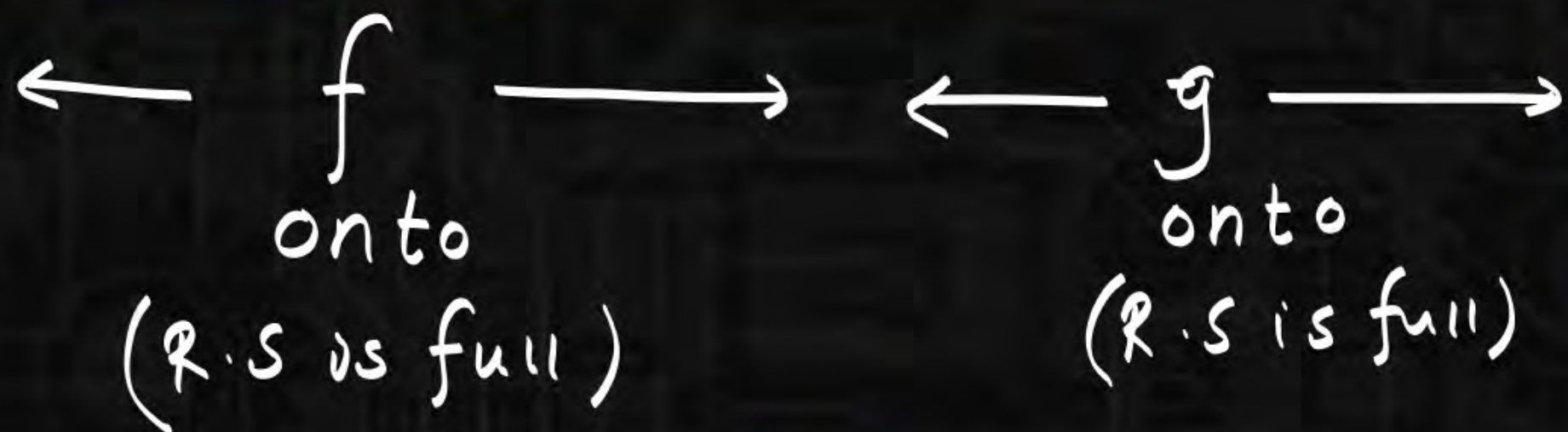
if f & g are onto then $g \circ f$ is onto. (True) ✓



Functions

- 1) if f & g are 1:1 then $g \circ f$ is 1:1. ✓
- 2) if f & g are onto then $g \circ f$ is onto. ✓
- 3) if f & g 1:1 then $g \circ f$ is 1:1. ✓

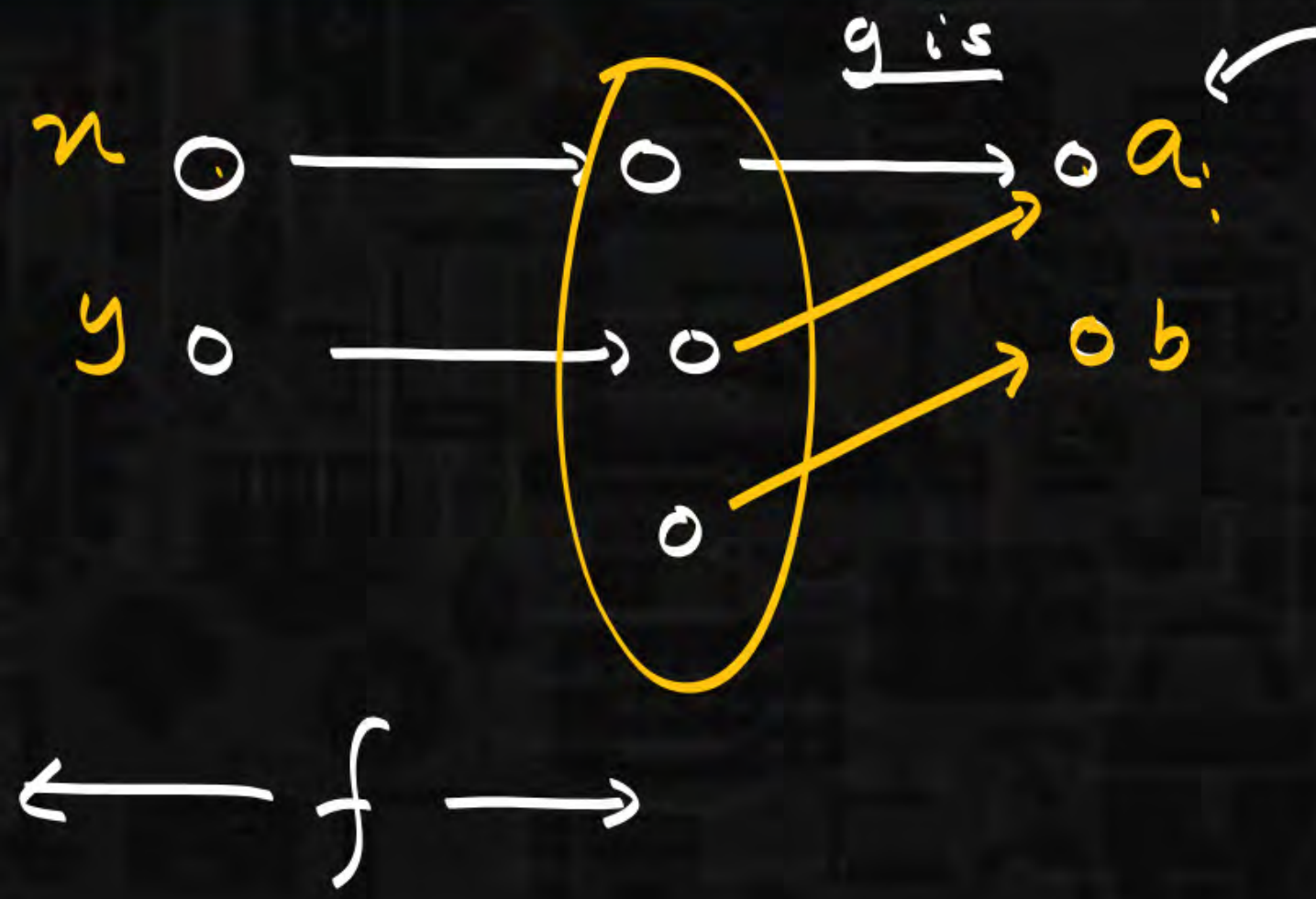
Functions



if f, g are onto then $g \circ f$ is onto.

if g is onto
then $g \circ f$ is onto.

Functions



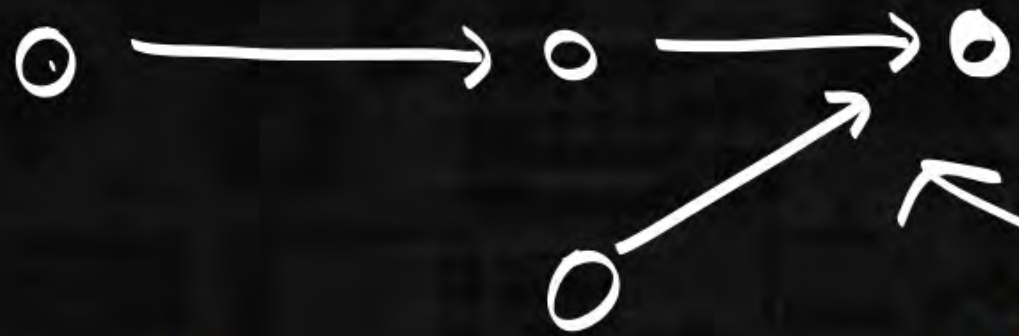
$g: \text{onto} \rightarrow g \circ f \text{ is onto.}$
(false)

if f, g are onto then $g \circ f$ is onto.

Functions



$g \circ f$ is onto $\longrightarrow$ g is onto
(True)



g is onto
 $\longrightarrow$ $g \circ f$ is onto
false.

$g \circ f$ is onto then f is onto (false)

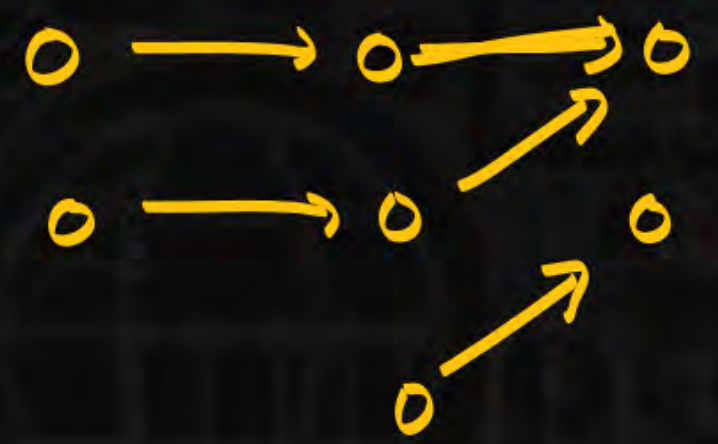
Functions

$g \circ f$ is onto then g is onto (True)



$g \circ f$ is onto then f is onto (False)

g is onto $\rightarrow g \circ f$ is onto (False)



Functions

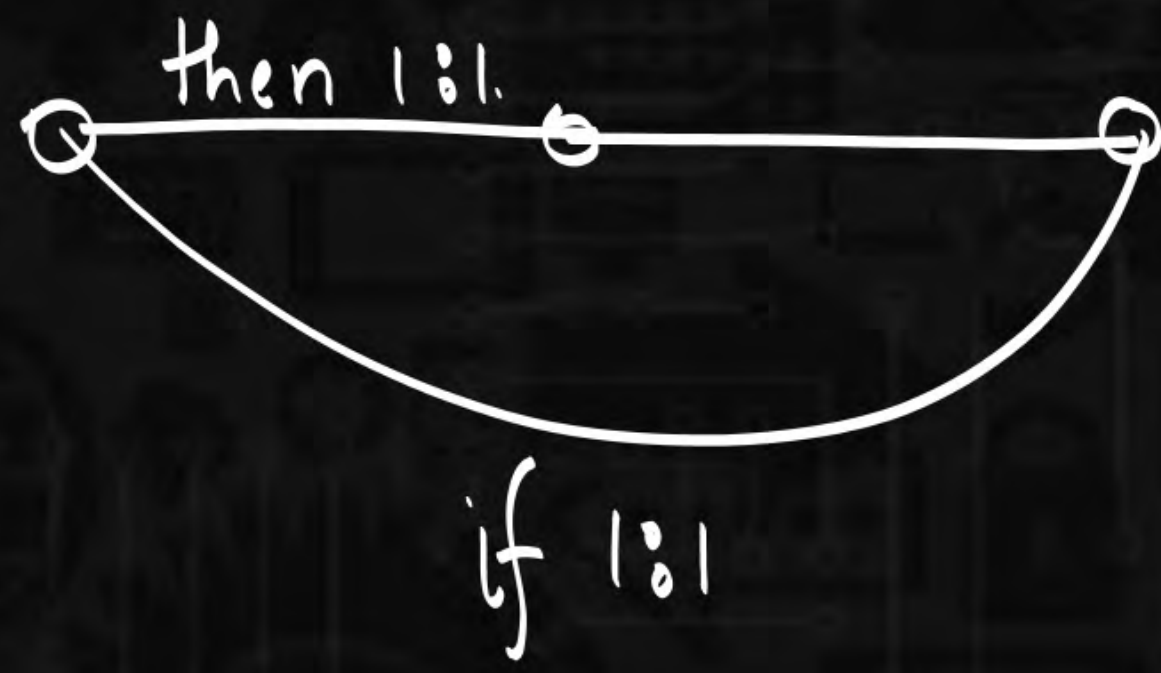
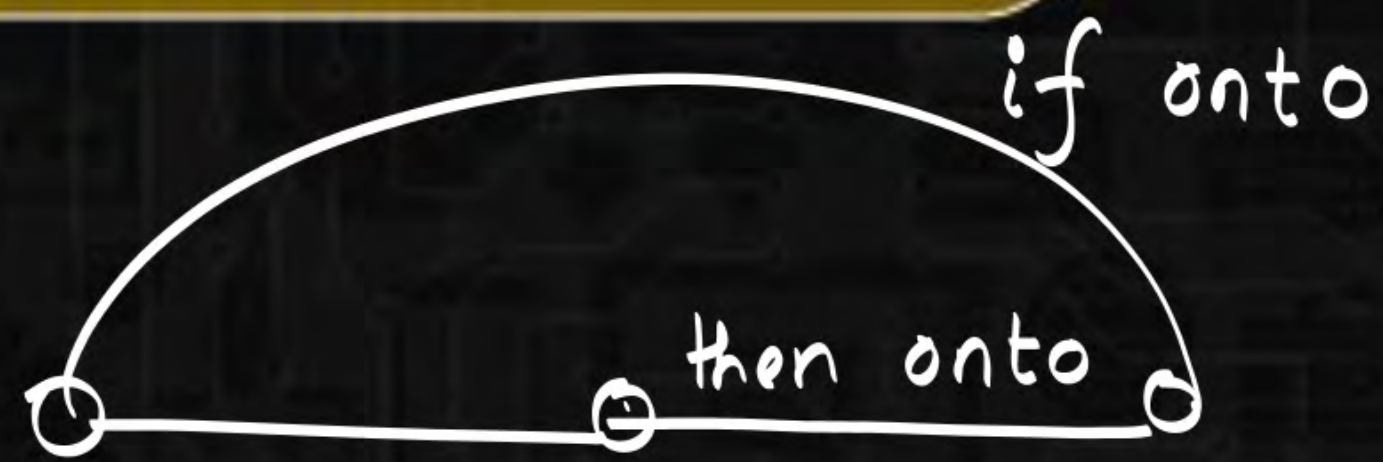
$g \circ f$ is 1:1 then g is 1:1 (false)



$g \circ f$ is 1:1 then f is 1:1 (True)



Functions



Functions



$$h(a) = g(f(a))$$

g is onto $\rightarrow h$ is onto.

h is onto $\rightarrow f$ is onto.

h is onto $\rightarrow g$ is onto.

h is onto $\rightarrow f$ & g are onto.

